An application begins either with two parentheses or with a parenthesis followed by a variable

The Open Logic Project contributors

openlogic:lambda-calculus/syntax:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-lambda-calculus-syntax/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the complete proof sentence with its citation to the term-start lemma is verbatim in the span.

Editorial note. The label lem:app-start names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Allowed standard prerequisites (editorial).

- Terms begin with a variable or a parenthesis. Same unit, \ollabel{lem:term-start} (source line 1541). Applied to the left argument P to split the two displayed openings.

Source reference. Original lemma lem-term-start: sources/olp-lambda-calculus-syntax/pinned-original-source.tex, source lines 1541–1541. Prerequisite pointer invoked at proof line(s) 1561; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *The result of an application starts with either two parentheses or a parenthesis and a variable.*

Proof. If M is the result of an application, it is of the form (PQ), so it begins with a parenthesis. Since P is a term, by lem-term-start, it begins either with a parenthesis or a variable. □